

PANDAS: TASK-01

Manipulation of Tips Data

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1.Introduction to Pandas Task:

```
#1.  
import pandas as pd  
import matplotlib.pyplot as plt
```

The dataset contains the information about the billing structure and customer information of the restaurant.

2. Loading and Understanding Data Task:

The dataset contains the details of a collection of customer bills from a restaurant. `total_bill` gives the total amount spent by a particular customer. `tip` column is the tip that is given by the customer for the service provided. Sex has two categories male or female. Smoker is also a binary column which answers whether a customer is a smoker or not. Day and time shows us when the customer visited the restaurant. The size column represent the total number of people (party size) at the table.

3. Data Exploration Task:

- Use `head()`
- Use `describe()`

The columns `total_bill` and `tips` contain numerical values while columns such as `sex`, `day`, `time`, `size` contain categorical values. The statistical summary of the dataset is obtained.

4. Indexing & Selection Task:

- Select the first 5 rows and columns `total_bill`, `tip` using `iloc`

```
#4. indexing and selection  
#Select the first 5 rows and columns total_bill, tip using iloc  
tips.iloc[:5][["total_bill","tip"]]
```

	total_bill	tip
0	16.99	1.01
1	10.34	1.66
2	21.01	3.50
3	23.68	3.31
4	24.59	3.61

- Select the last 3 rows and columns `day`, `time` using `iloc`

```
#Select the last 3 rows and columns day, time using iloc
tips.iloc[-3:][["day", "time"]]
```

	day	time
241	Sat	Dinner
242	Sat	Dinner
243	Thur	Dinner

- Select rows with index labels 0 to 10 and columns total_bill, sex using loc

```
#Select rows with index labels 0 to 10 and columns total_bill, sex using
tips.loc[0:10, ["total_bill", "sex"]]
```

	total_bill	sex
0	16.99	Female
1	10.34	Male
2	21.01	Male
3	23.68	Male
4	24.59	Female
5	25.29	Male
6	8.77	Male
7	26.88	Male
8	15.04	Male
9	14.78	Male
10	10.27	Male

- Select only the tip column for all rows using loc

```
#Select only the tip column for all rows using loc
tips.loc[:, "tip"]
```

	tip
0	1.01
1	1.66
2	3.50
3	3.31
4	3.61
...	...
239	5.92
240	2.00
241	2.00
242	1.75
243	3.00

244 rows × 1 columns

dtype: float64

- Select row with index 5 and columns total_bill, size using loc

```
#Select row with index 5 and columns total_bill, size using loc
tips.loc[5, ["total_bill", "size"]]
```

	5
total_bill	25.29
size	4

dtype: object

- Select rows from position 10 to 20 and columns from position 1 to 3 using iloc

```
#Select rows from position 10 to 20 and columns from position 1 to 3 using
tips.iloc[10:21,1:4]
```

	tip	sex	smoker
10	1.71	Male	No
11	5.00	Female	No
12	1.57	Male	No
13	3.00	Male	No
14	3.02	Female	No
15	3.92	Male	No
16	1.67	Female	No
17	3.71	Male	No
18	3.50	Female	No
19	3.35	Male	No
20	4.08	Male	No

- Select the first 10 rows and first 2 columns using iloc

```
#Select the first 10 rows and first 2 columns using iloc
tips.iloc[:10,:2]
```

	total_bill	tip
0	16.99	1.01
1	10.34	1.66
2	21.01	3.50
3	23.68	3.31
4	24.59	3.61
5	25.29	4.71
6	8.77	2.00
7	26.88	3.12
8	15.04	1.96
9	14.78	3.23

- Select rows where total_bill > 20 and columns total_bill, tip using loc

```
#Select rows where total_bill > 20 and columns total_bill, tip using loc
tips.loc[tips["total_bill"]>20,["total_bill","tip"]]
```

	total_bill	tip
2	21.01	3.50
3	23.68	3.31
4	24.59	3.61
5	25.29	4.71
7	26.88	3.12
...	...	...
237	32.83	1.17
238	35.83	4.67
239	29.03	5.92
240	27.18	2.00
241	22.67	2.00

- Select rows where day == 'Sunday' and columns total_bill, tip, size using loc

```
#Select rows where day == 'Sunday' and columns total_bill, tip, size using
tips.loc[tips["day"]=="Sun",["total_bill","tip","size"]]
```

	total_bill	tip	size
0	16.99	1.01	2
1	10.34	1.66	3
2	21.01	3.50	3
3	23.68	3.31	2
4	24.59	3.61	4
...	...	...	...
186	20.90	3.50	3
187	30.46	2.00	5
188	18.15	3.50	3
189	23.10	4.00	3
190	15.69	1.50	2

5. Filtering Data Task:

- Filter high bill customers
- Filter by gender
- Filter by time Insight

Customers with total bills greater than 30 are fewer in number but generally leave higher tip amounts. Male customers appear more frequently in the dataset than female customers. Dinner records are more common than lunch records, indicating that dinner is the busier service period. Customers dining during dinner tend to have higher bills and tips compared to lunch customers. Filtering helps identify specific customer segments for targeted analysis, such as high-spending customers or dinner visitors.

6. Value Counts & Distribution Task:

- Use `value_counts()`
- Use `normalize=True` Insight

```
tips["sex"].value_counts()
```

count

sex

Male	157
Female	87

dtype: int64

```
tips["sex"].value_counts(normalize=True)
```

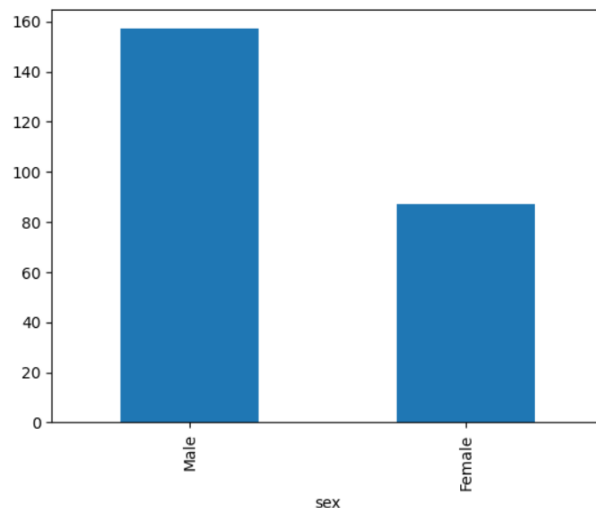
proportion

sex

Male	0.643443
Female	0.356557

dtype: float64

<Axes: xlabel='sex'>



The number of males and females coming to the restaurant differs significantly. The males are more.

7. Sorting Data Task

- Sort by total_bill
- Sort by tip

the sorting is done according to the total bill and tip and we found that highest bills usually receive higher tips. It was also found that higher the bill higher will be the tip

8. Handling Missing Values Task:

- Check missing values
- Apply handling if needed

There were no missing values

9. Feature Creation Task:

- Create a new feature tip_percentage, which represents the tip as a percentage of the total bill. This can be calculated by dividing the tip by total_bill and multiplying by 100. This feature helps in understanding tipping behavior relative to the bill amount.

```
#9.Feature Creation
#Tip Percentage
tips["tip_percentage"]=(tips["tip"]/tips["total_bill"])*100
```

```
tips.head()
```

	total_bill	tip	sex	smoker	day	time	size	tip_percentage
0	16.99	1.01	Female	No	Sun	Dinner	2	5.944673
1	10.34	1.66	Male	No	Sun	Dinner	3	16.054159
2	21.01	3.50	Male	No	Sun	Dinner	3	16.658734
3	23.68	3.31	Male	No	Sun	Dinner	2	13.978041
4	24.59	3.61	Female	No	Sun	Dinner	4	14.680765

Next steps: [Generate code with tips](#) [New interactive sheet](#)

Through this we can find out that the tips lie somewhere between 5-20 percentage of the bill.

- Create a scenario using any column so that you can apply cut method in pandas to add bins to the data.

```
#Binning
tips["Bill_Category"]=pd.cut(
    tips["total_bill"],
    bins=[0,10,20,30,60],
    labels=["Low","Medium","High","Very High"]
)
```

```
tips.head()
```

tip	sex	smoker	day	time	size	tip_percentage	Bill_Category
1.01	Female	No	Sun	Dinner	2	5.944673	Medium
1.66	Male	No	Sun	Dinner	3	16.054159	Medium
3.50	Male	No	Sun	Dinner	3	16.658734	High
3.31	Male	No	Sun	Dinner	2	13.978041	High
3.61	Female	No	Sun	Dinner	4	14.680765	High

10. GroupBy Analysis Task:

- Group by gender
- Group by day
- Group by time

Dinner customers spend more. Saturday has higher average bills. Average tip differs slightly by gender.

11. Multi-Level Grouping Task:

- Group by multiple columns of your choice

```
#11.Multi-Level Grouping
tips.groupby(["day","sex"]).mean(numeric_only=True)
```

		total_bill	tip	size	tip_percentage
day	sex				
Fri	Female	14.145556	2.781111	2.111111	19.938840
	Male	19.857000	2.693000	2.100000	14.338519
Sat	Female	19.680357	2.801786	2.250000	15.647021
	Male	20.802542	3.083898	2.644068	15.157684
Sun	Female	19.872222	3.367222	2.944444	18.156877
	Male	21.887241	3.220345	2.810345	16.234407
Thur	Female	16.715312	2.575625	2.468750	15.752483
	Male	18.714667	2.980333	2.433333	16.527649

Spending varies according to both gender and day.

12. Crosstab Analysis Task:

- Create crosstab of your own choice

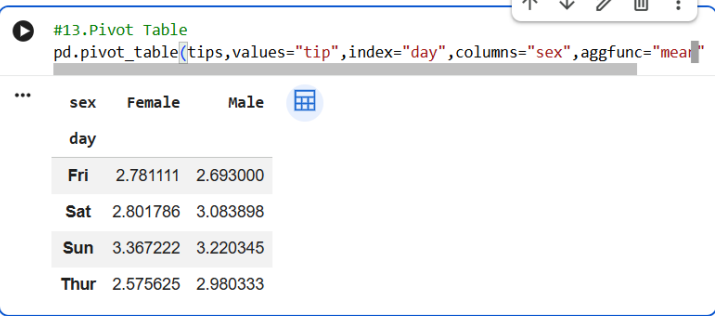
```
#12.Crosstab
pd.crosstab([tips["day"],tips["time"]])
```

time	Dinner	Lunch
day		
Fri	12	7
Sat	87	0
Sun	76	0
Thur	1	61

Dinner is more common than lunch. Especially during weekends people come more frequently to eat dinner.

13. Pivot Tables Task:

- Create pivot table of your own choice



```
#13.Pivot Table
pd.pivot_table(tips, values="tip", index="day", columns="sex", aggfunc="mean")
```

sex	Female	Male
day		
Fri	2.781111	2.693000
Sat	2.801786	3.083898
Sun	3.367222	3.220345
Thur	2.575625	2.980333

Male customers leave higher average tips than female customers on both Saturday and Thursday, while female customers leave slightly higher average tips on Friday and Sunday. Sunday has the highest average tip for both genders, indicating that customers tend to tip more on that day. The differences in average tips between males and females are relatively small, suggesting that the day of the visit has a greater influence on tipping than gender.

14. Ranking Data Task:

- Rank based on bill
- Rank based on tip percentage

Customers with the same bill amount or tip percentage receive the same average rank (for example, ranks such as 82.5 or 219.5), showing that the rank() function assigns tied values the average of their positions. Customers with higher total bills generally receive better (lower-numbered) bill ranks, making it easy to identify the restaurant's highest spenders. The Tip_Rank is based on tip percentage rather than tip amount, so customers with smaller bills can still achieve a high rank if they leave a generous percentage as a tip. A customer with a high Bill_Rank does not always have a high Tip_Rank, indicating that spending more does not necessarily mean tipping a higher percentage

15. Value Transformation Task:

- Use map()

A new column is created where males are represented as M and females are represented as F

16. Category Understanding Task:

- Use unique()
- Use nunique()

There were 4 unique categories under the day column which are Sun, Sat, Thur, Fri

17. Handling Duplicates Task:

- Check duplicates
- Remove duplicates.

A duplicate value was found and it was handled by removing the entire row.

18. Column Manipulation Task:

- Rename columns of your choice

```
#18. Column Manipulation
#Rename Columns
tips.rename(columns={
    "total_bill":"Bill",
    "tip":"Tip"
},inplace=True)
```

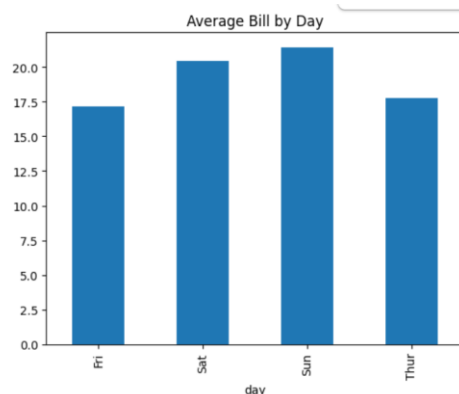
```
tips.head()
```

	Bill	Tip	sex	smoker	day	time	size	tip_percentage	Bill_C
0	16.99	1.01	Female	No	Sun	Dinner	2	5.944673	
1	10.34	1.66	Male	No	Sun	Dinner	3	16.054159	
2	21.01	3.50	Male	No	Sun	Dinner	3	16.658734	
3	23.68	3.31	Male	No	Sun	Dinner	2	13.978041	
4	24.59	3.61	Female	No	Sun	Dinner	4	14.680765	

19. Data Visualization using pandas Task:

- Bar chart

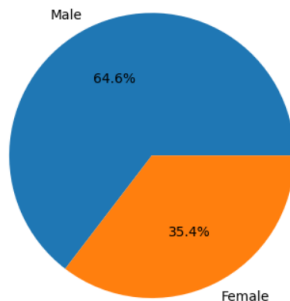
```
#19.Visualization
#bar chart
tips.groupby("day")["Bill"].mean().plot(kind="bar")
plt.title("Average Bill by Day")
plt.show()
```



People spend more on Sunday as compared to other days. weekend shows a higher average spending as compared to weekdays.

- Pie chart

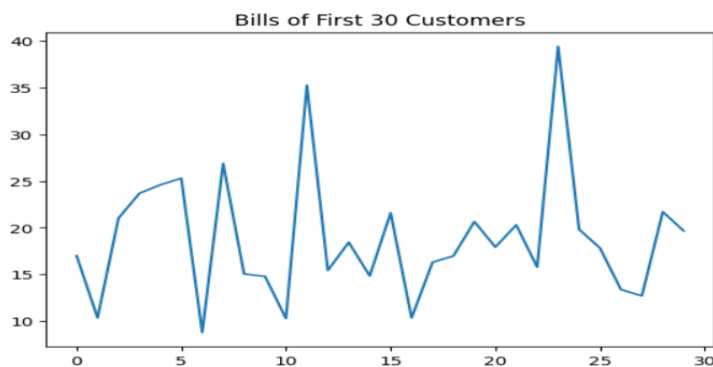
```
#pie chart
tips["sex"].value_counts().plot(kind="pie", autopct="%1.1f%%")
plt.ylabel("")
plt.show()
```



Male customers make up the majority of the restaurant's customers, accounting for approximately 64.6% of the dataset, while female customers account for about 35.4%. The proportion of male customers is almost twice that of female customers, indicating that the dataset contains significantly more male dining records.

- Line chart

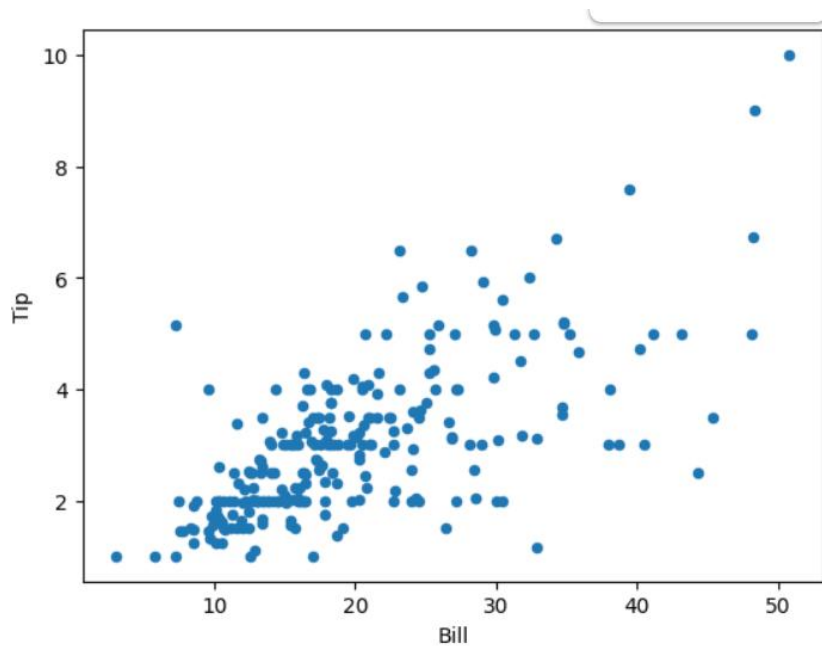
```
#line chart
tips["Bill"].head(30).plot(kind="line")
plt.title("Bills of First 30 Customers")
plt.show()
```



The bill shows a huge fluctuation across the taken 30 customers. Some customers spend more while others spend less. There is no consistent increase or decrease seen.

- Scatter plot

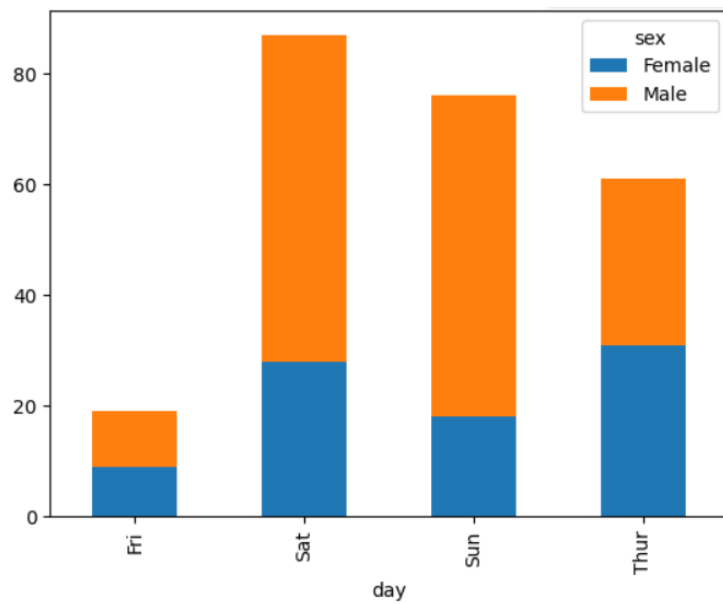
```
#Scatter Plot
tips.plot(
    kind="scatter",
    x="Bill",
    y="Tip"
)
plt.show()
```



There is a positive relationship between the total bill and the tip amount, indicating that customers with higher bills generally leave higher tips. Most customer bills fall between approximately 10 and 30, with corresponding tips ranging from about 1 to 5, showing that the majority of transactions are in the moderate spending range. Although the overall trend is positive, customers with similar bill amounts sometimes leave different tip amounts, suggesting that tipping behavior varies from customer to customer.

- Stacked bar chart

```
#Stacked Bar Chart
pd.crosstab(
    tips["day"],
    tips["sex"]
).plot(kind="bar", stacked=True)
plt.show()
```



We can conclude from this visualization that Saturday is when the number of customers is high and Friday is when the number of customers is low. During weekend male customers visit the restaurant more than female customers. The restaurant is seen to be busier in the weekends.